MolPallete — Corpus EDA

naveja_recap

/home/mogan/preprocessed/molpallete/flavordb-only/naveja_recap

generated 2026-08-31 12:29

A · Provenance

field	value
sources	['flavordb']
method	naveja_recap
decomposition_params	{'ratio': 0.3333333333333333, 'include_ring': True, 'max_cores':
core_ratio	0.3333333333333333
max_cores	4
max_rgroups	8
keep_stereo	True
neutralise	False
dedup	True
n_duplicates_removed	1615
size_filter	{'min_heavy_atoms': 5, 'max_heavy_atoms': 50}
graph_hash_version	4
rdkit_version	2026.03.2
created_at	2026-08-31T09:25:14

B · Composition

quantity	value
molecules	20,812
from flavordb	20,812 (100.0%)
decompositions	65,901
per molecule	mean 3.17, median 4, max 4
R-group slots	79,057
per decomposition	mean 1.20
k >= 2	16.58% of decompositions

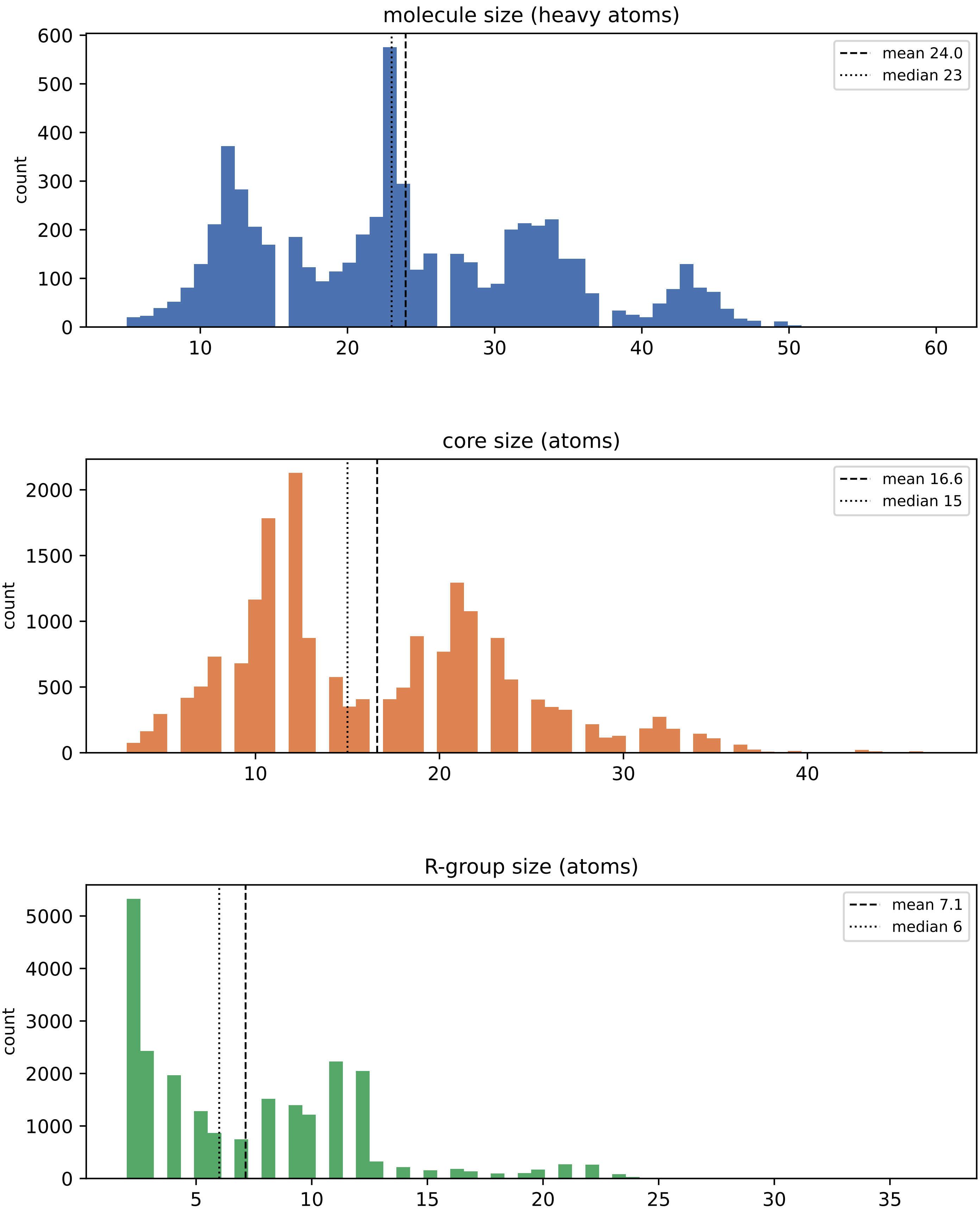
k>=2 is what the islinked subset space and the core-decoration objective have to work with.

C · Augmentation space

scheme	size
sampling_unit = molecule	20,812 instances/epoch
sampling_unit = decomposition	65,901 instances/epoch (3.2x)
full (mol, core, islinked) space	101,205 (4.9 per molecule)

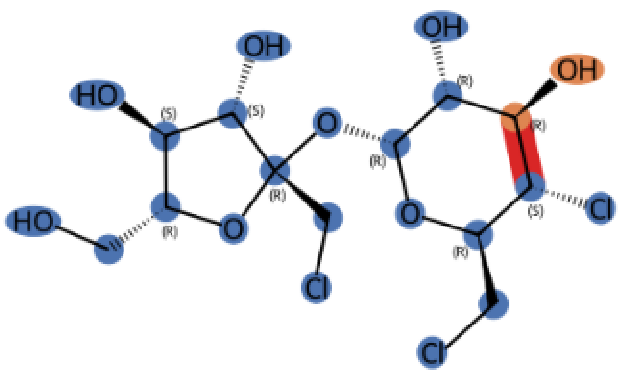
MolPLA's two stages: one molecule yields many cores; one core yields 2^k - 1 islinked subsets.

D · Size distributions

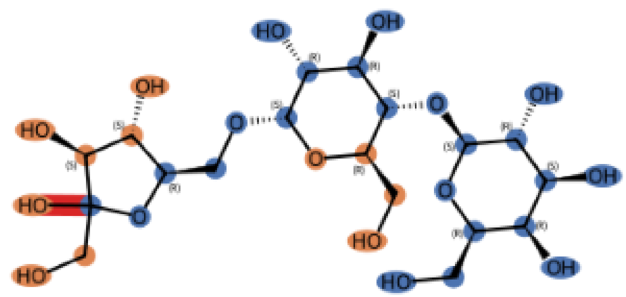


E · Example decompositions (ratio 0.3333333333333333)

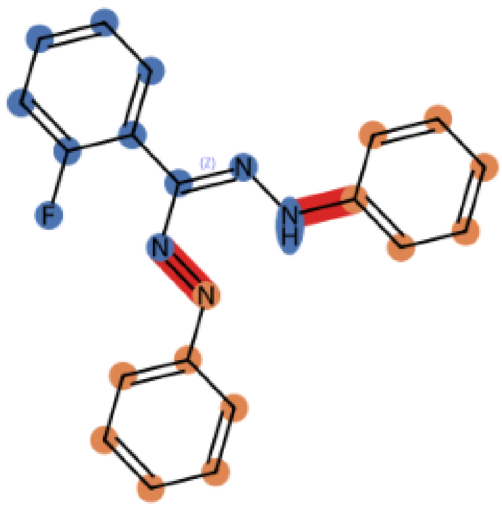
FDB10000891   k=1   core 21 atoms, R-groups [2]



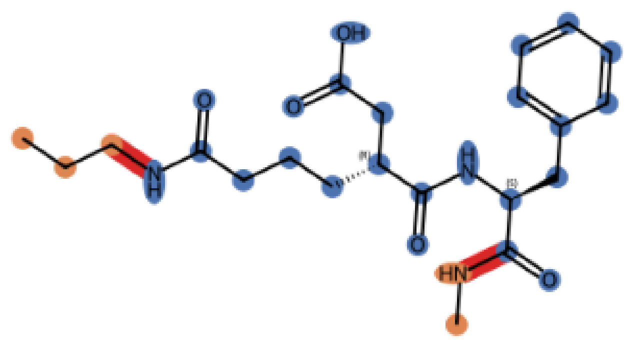
FDB10006309   k=1   core 23 atoms, R-groups [11]



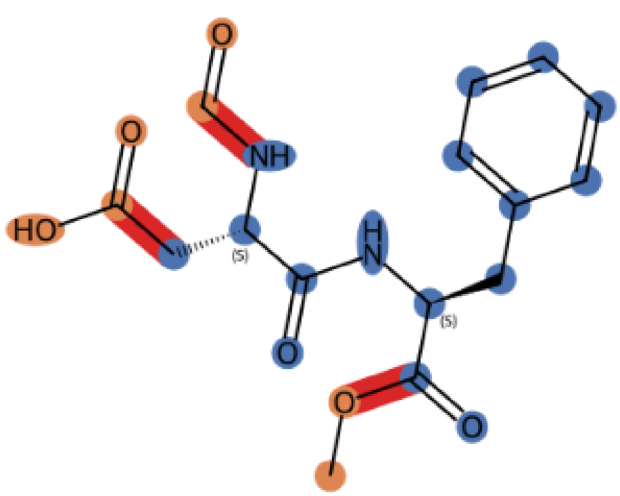
FDB100081134   k=2   core 11 atoms, R-groups [7, 6]



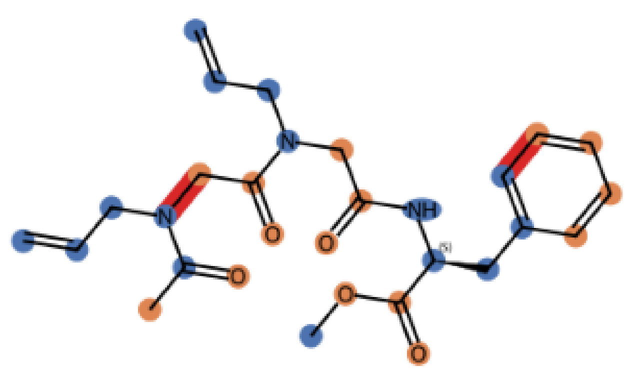
FDB10024256   k=2   core 24 atoms, R-groups [3, 2]



FDB10268056   k=3   core 16 atoms, R-groups [2, 3, 2]

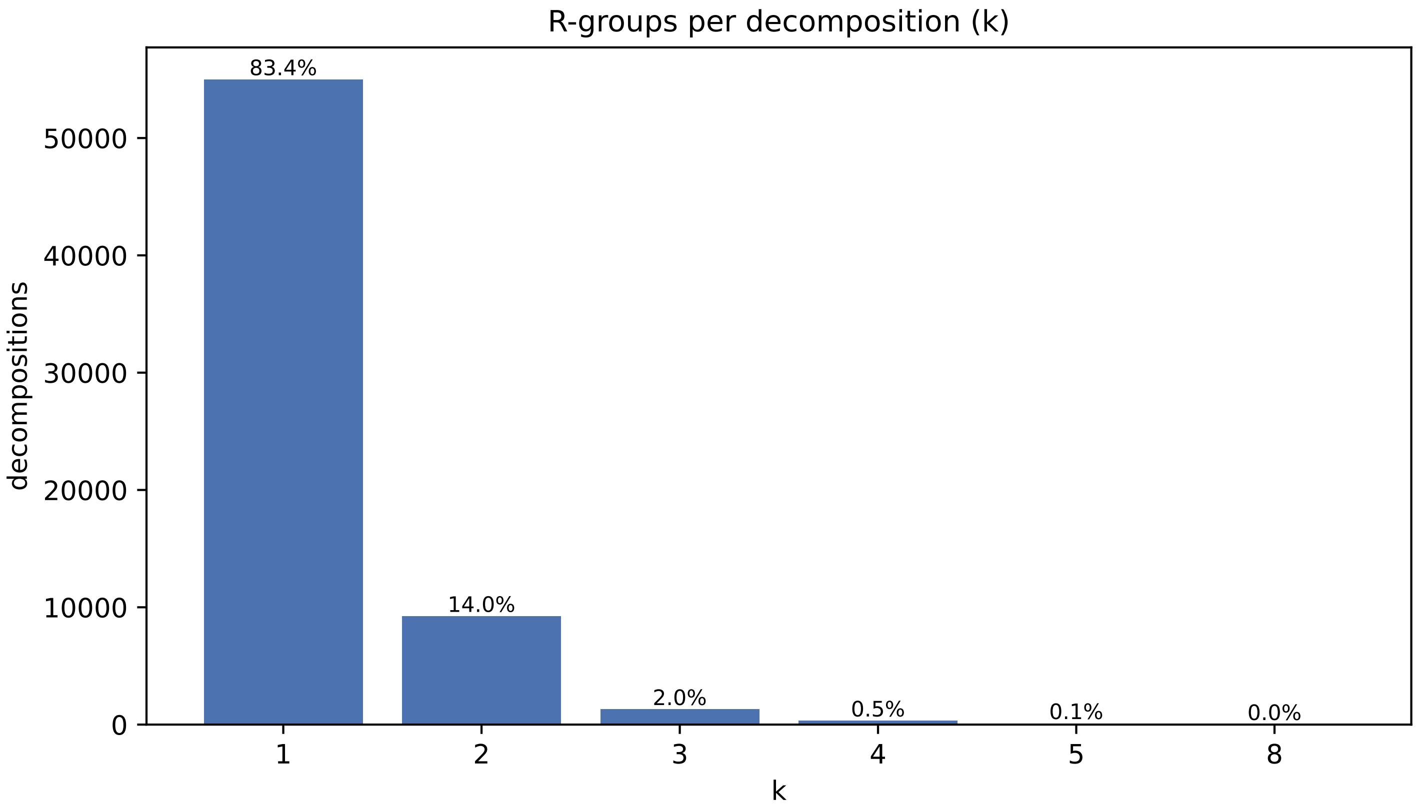


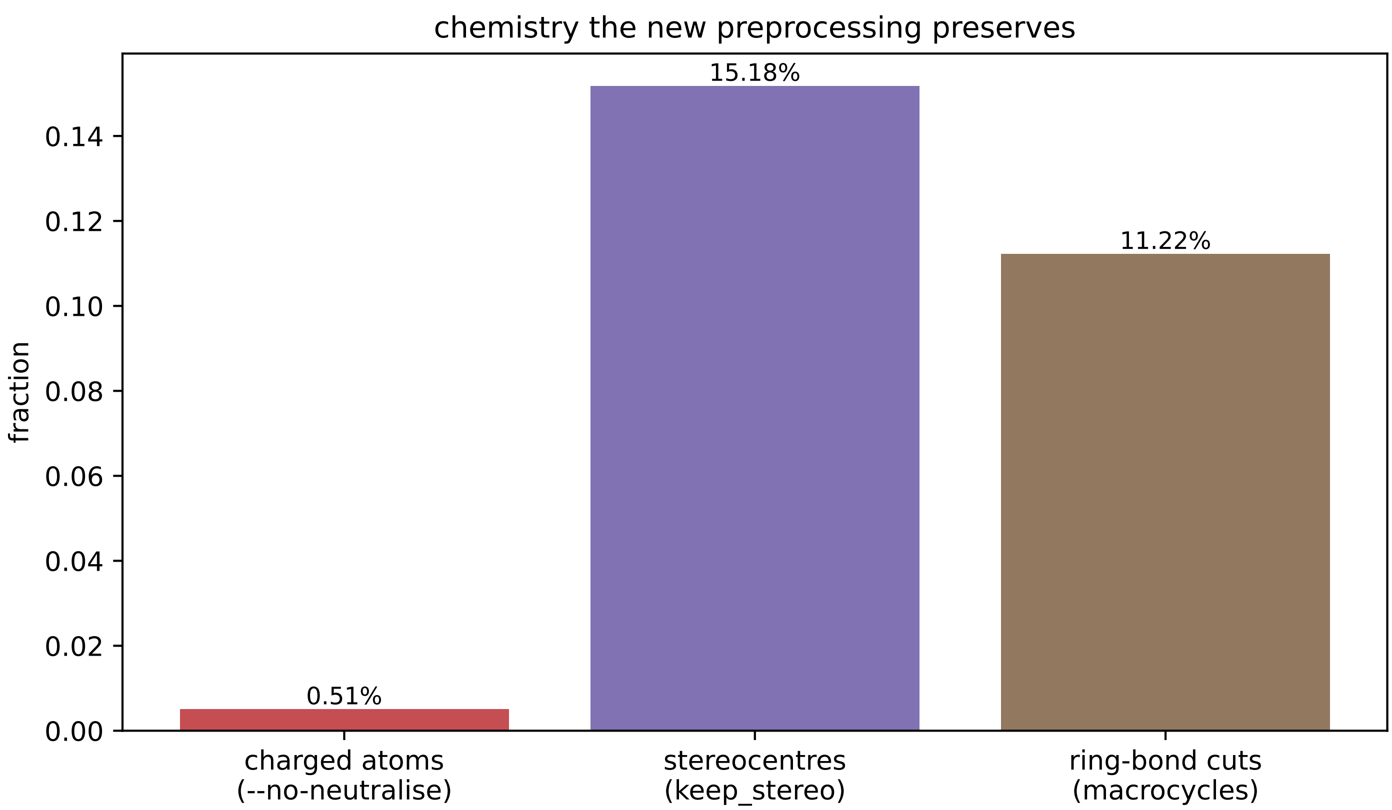
FDB10573892   k=4   core 15 atoms, R-groups [3, 3, 7, 2]



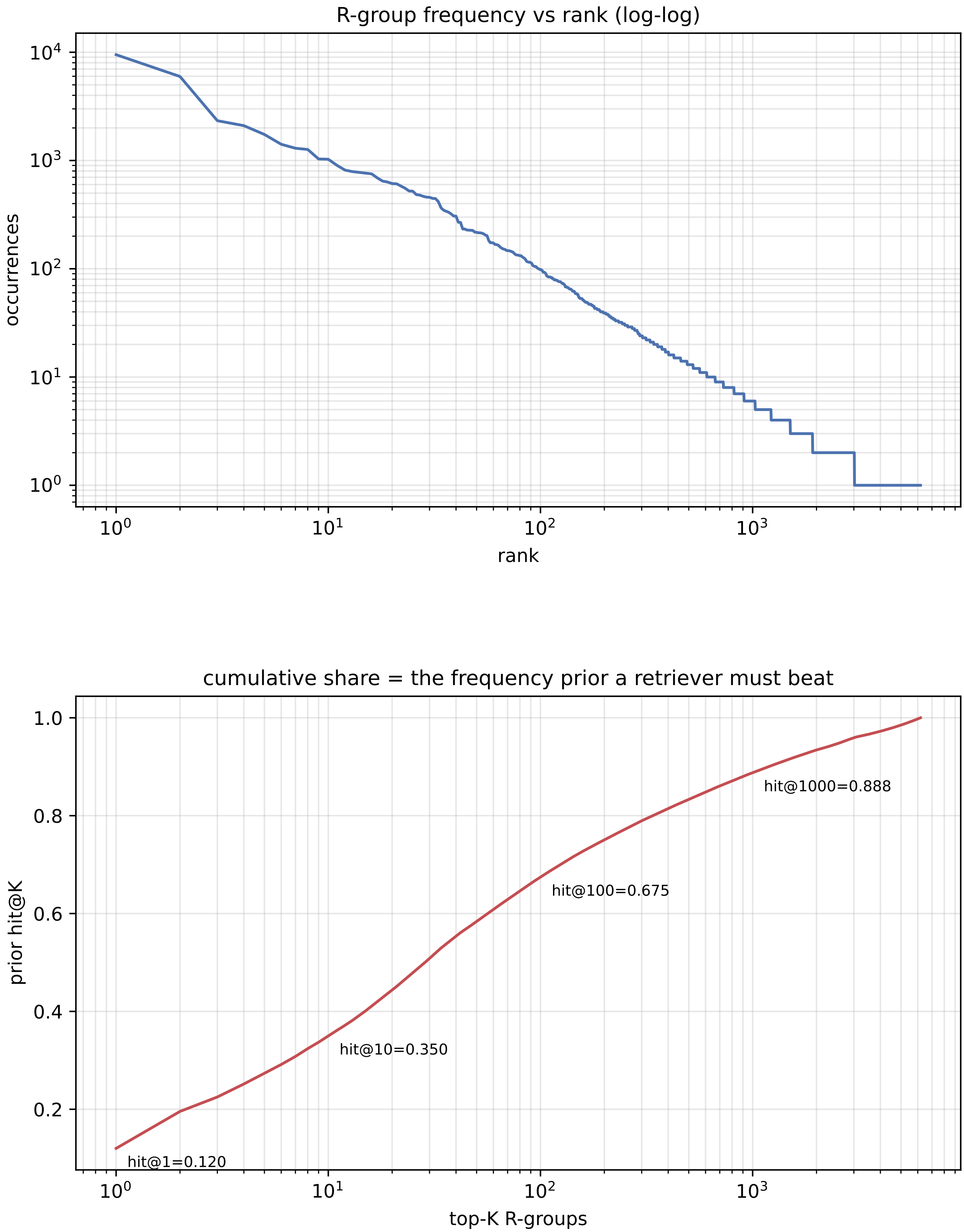
blue = core   orange = R-groups   red = cut bonds

F · Decomposition shape & preserved chemistry





G · R-group library skew



H · R-group library

quantity	value
distinct R-groups	6,182
occurrences	79,057
effective size (exp H)	299
top-1 share	12.01%
top-10 cover	35.0%
top-100 cover	67.5%

Effective size, not the raw count, is the number of classes retrieval actually chooses between.

I · Most frequent R-groups

count	share	SMILES
9,495	12.01%	*[CH2]O
5,971	7.55%	*OC
2,332	2.95%	*O[C@@H]1O[C@H](CO)[C@@H](O)[C@H](O)[C
2,099	2.66%	*C(C)=O
1,740	2.20%	*[CH2]C
1,412	1.79%	*[NH]C(Cc1ccccc1)C(=O)O
1,298	1.64%	*C1O[C@H](CO)[C@@H](O)[C@H](O)[C@@H]1O
1,265	1.60%	*c1ccc(O)c(O)c1
1,033	1.31%	*P(=O)(O)O
1,027	1.30%	*O[C@@H]1O[C@@H](C)[C@H](O)[C@@H](O)[C
901	1.14%	*OP(=O)(O)O
815	1.03%	*n1cnc2c(N)ncnc21
790	1.00%	*C(=O)C(N)CC(=O)O
777	0.98%	*C(O)CO
763	0.97%	*[CH]=O
753	0.95%	*c1ccc(O)c(OC)c1
692	0.88%	*[CH2]CC
646	0.82%	*C(=O)O
633	0.80%	*[CH2]CO
612	0.77%	*OCC

J · Instance-level common-R-group filter

quantity	value
percentile	99.99
count threshold	7,317
hashes flagged common	1 of 6,182
their share of occurrences	12.0%
single-R-group instances rejected	2,435/23,031 (10.6%)
mean R-group size, rejected	2.00 atoms
mean R-group size, kept	7.75 atoms

Filters INSTANCES, not the library: the retrieval target space and its frequency prior are unchanged. At k=1 the rule reduces to 'drop if the R-group is common', so rejection is far heavier than MolPLA saw.